

DeepECC: a deep learning framework for genome-wide identification and analysis of human cancer eccDNAs

Changcheng Wang¹, Yisen Xu¹, Rufeng Li¹, Min Qiang¹, Chen Guo¹, Qian He², Xiaokang Wu², Lingmi Hou³, Qiuran Xu^{4,*}, Yungang Xu^{1,2,*}

¹Department of Cell Biology and Genetics, School of Basic Medical Sciences, Xi'an Jiaotong University Health Science Center, Xi'an, Shaanxi 710061, China

²Department of Clinical Laboratories, The Second Affiliated Hospital of Xi'an Jiaotong University, Xi'an, Shaanxi 710004, China

³Breast Surgery, Sichuan Cancer Hospital and Institute, Sichuan Cancer Center, School of Medicine, University of Electronic Science and Technology of China, Chengdu, Sichuan 611731, China

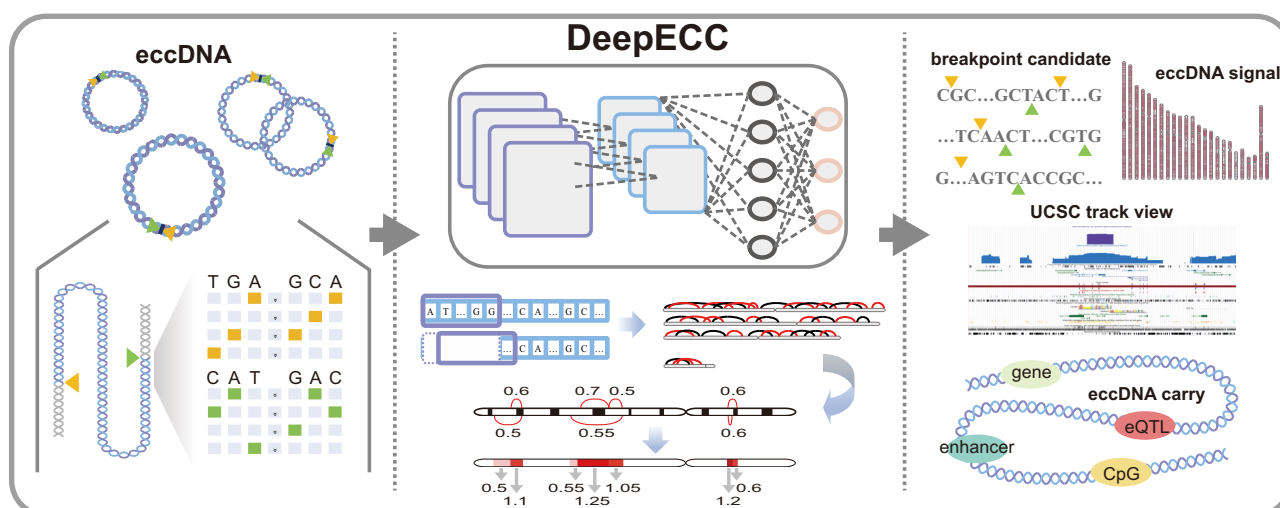
⁴Zhejiang Key Laboratory of Tumor Molecular Diagnosis and Individualized Medicine, Zhejiang Provincial People's Hospital, Affiliated People's Hospital, Hangzhou Medical College, Hangzhou, Zhejiang 310014, China

*To whom correspondence should be addressed. Email: Yungang.xu@xjtu.edu.cn
 Correspondence may also be addressed to Qiuran Xu. Email: xuqiuran@hmc.edu.cn

Abstract

Extrachromosomal circular DNAs (eccDNAs) are closed circular DNA molecules widespread across eukaryotic cells, with emerging roles in gene regulation and tumor progression. Experimental assays remain costly and incomplete, underscoring the need for computational approaches. To address this, a deep learning framework termed DeepECC has been established to overcome the challenges posed by eccDNA heterogeneity and its complex biogenesis. Through a two-stage training strategy, DeepECC models the local sequence context flanking both the start and end breakpoints, thereby capturing mechanistically informative features that are often overlooked when analyses focus solely on eccDNA body sequences. Applied to multi-species (human, mouse, gallus) datasets, DeepECC robustly captures conserved breakpoint features, with a marked preference for GC-rich and transcriptionally active regions. Genome-wide scanning reveals non-uniform distributions of human cancer eccDNAs enriched in enhancers, expression quantitative trait loci, and CTCF sites, suggesting regulatory functions in tumor progression. Motif analysis further implicates ribosomal activity, translational regulation, and DNA damage response. Furthermore, genome-wide eccDNA predictions are integrated into the UCSC Genome Browser, enabling convenient querying and visualization of cancer-related eccDNAs associated with specific genes or genomic regions, facilitating functional interpretation for experimental research. Collectively, DeepECC provides a generalizable framework for systematic eccDNA discovery and insights into their functional significance in cancer.

Graphical abstract



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Introduction

Extrachromosomal circular DNA (eccDNA), a type of closed circular DNA independent of chromosomes, has recently attracted increasing attention in the field of genome biology. These molecules exhibit distinctive characteristics, including autonomous replication, independence from cell cycle regulation [1], and widespread occurrence across eukaryotes from normal tissues to disease-associated cells [2]. EccDNAs are highly heterogeneous in origin and size, ranging from dozens of base pairs to megabases, and can be classified into subtypes such as extrachromosomal DNA (ecDNA), extrachromosomal rDNA circle (ERC), t-circles, spcDNA, and microDNA [3]. Their biogenesis involves multiple mechanisms, including homologous recombination, non-homologous end joining, chromosomal breakage and self-ligation, replication-associated errors, and R-loop formation [4–6]. Functionally, eccDNAs are far from passive byproducts of genomic instability but have diverse effects on cell growth and function [7, 8]. Large ecDNAs often carry genes and drive their overexpression, particularly in cancer [9–11], while smaller eccDNAs may contain enhancer fragments or other regulatory elements that modulate transcription in *cis* or *trans* [12–14]. Owing to their chromosomal origin and persistence within cells [15, 16], eccDNAs have been implicated in processes such as drug resistance, immune modulation, and adaptive evolution [17, 18]. Collectively, these findings suggest that eccDNAs are not merely structural artifacts of genome rearrangement but active regulators of gene function and cellular behavior in both normal and disease contexts.

Currently, the identification of eccDNA mainly relies on sequencing-based technologies, such as DNA-seq, ATAC-seq, and Circle-seq [19, 20], followed by computational tools (e.g. AmpliconReconstructor [21], Circle_finder [22], and ecc_finder [23]) that align sequencing reads to the reference genome and identify junction sites to reconstruct the circular DNA structure. These advances, together with specialized databases including CircleBase [24], eccDNA Atlas [25], and TeCD [26], have expanded our understanding of ecDNA diversity and distribution. However, experimental and alignment-based approaches remain costly, labor-intensive, and often fail to capture low-abundance or structurally complex eccDNAs, underscoring the need for efficient computational strategies enabling genome-wide identification and analysis. Therefore, recent studies have explored deep learning for sequence-driven eccDNA prediction [27, 28]. DeepCircle, for instance, employs convolutional neural networks (CNNs) and BERT architectures trained on fixed 1000 bp sequences from the eccDNA body to predict whether a DNA segment represents an eccDNA element [28]. DeepCircle shows promise but is constrained by two major limitations. First, it relies on fixed 1000 bp sequences from the eccDNA body for training and prediction, which requires excluding eccDNAs longer than 1000 bp and padding shorter ones, potentially discarding important signals or introducing noise, and therefore cannot accommodate the extensive length heterogeneity of eccDNAs, which ranges from tens of base pairs to several megabases. Second, from the biological perspective, sequences flanking eccDNA breakpoints are likely more informative for predicting eccDNA formation than the internal sequences within the eccDNA body.

To address the above challenges, we developed DeepECC, a novel deep learning framework designed to learn the in-

trinsic sequence features surrounding eccDNA breakpoints. Unlike existing eccDNA reconstruction tools such as Circle_finder and AmpliconReconstructor, which infer circular DNA structures from sequencing read alignments, as well as learning-based methods that model eccDNA body sequences, DeepECC focuses on the sequence features flanking eccDNA start and end breakpoints. By integrating the internal sequences adjacent to breakpoints with the surrounding external genomic context, DeepECC is better positioned to capture the underlying mechanisms of eccDNA formation and to provide stronger discriminative power. DeepECC employs a two-stage framework for genome-wide identification: the first stage scans the genome to identify candidate breakpoints, while the second stage evaluates potential start–end pairs to infer their likelihood of forming closed circular structures, thereby generating a genome-wide probabilistic landscape of eccDNA formation. By applying DeepECC to multiple human and cross-species datasets, we demonstrated that the model robustly learns conserved breakpoint features across chromosomes and species, with a marked preference for GC-rich and transcriptionally active regions. Genome-wide analysis of human cancer eccDNA revealed non-uniform distributions of eccDNA signals, uncovering hotspots enriched in genes, regulatory elements, and cancer-related pathways. Furthermore, integrative analyses showed that predicted human cancer eccDNAs frequently overlap with enhancers, expression quantitative trait loci (eQTLs), CTCF binding sites, and other functional regulatory loci, highlighting their potential to influence chromatin architecture, transcriptional regulation, and tumor progression. Together, these results establish DeepECC as a powerful and generalizable framework for systematic genome-wide identification of eccDNAs. Beyond providing single-base resolution predictions, DeepECC offers new insights into the sequence determinants and functional relevance of eccDNA formation, shedding light on its regulatory and pathogenic roles in cancer and other biological contexts.

Materials and methods

Data sources

In this study, we utilized eccDNA datasets from three publicly available databases: CircleBase [24], eccDNA Atlas [25], and TeCD [26]. These databases curate experimentally validated eccDNAs reported in previously published studies and encompass a wide range of biological sources, including human cancer cells, human normal cells, mouse tissues, and the gallus species. For human eccDNA data, we extracted the genomic coordinates of the start and end positions of each eccDNA and mapped them uniformly to the human reference genome (hg38 version). For other species, eccDNA coordinates were mapped to mm10 and galGal6 reference genomes, respectively. Given the broad length distribution of eccDNAs, we applied minimal filtering during preprocessing, retaining eccDNAs with lengths between 50 bp and 50 Mb, and removed fully redundant records for downstream analysis and model training. Detailed statistics are provided in [Supplementary Table S1](#). Due to high redundancy among the databases, particularly between CircleBase and the others in cancer-related data, we primarily used human cancer cell eccDNAs from CircleBase for training the main predictive model to ensure data independence and consistency.

To comprehensively investigate the distribution and potential biological functions of eccDNAs, we also incorporated a variety of genomic feature annotations from multiple sources. The data types are summarized in [Supplementary Table S12](#). Specifically, we obtained 38 categories of genomic feature regions from the FusionAI project [29], including repetitive regions, structural variation regions, chromatin state regions, and regulatory regions. These annotations were originally based on the hg19 human genome assembly. To align them with the eccDNA breakpoint data, we used the LiftOver tool to convert all coordinates to the hg38 version for consistency. Enhancer region data were obtained from the SEDb database [30], which provides coordinates of super enhancers and typical enhancers identified across various cancer types. These data were used for overlap analysis with cancer-derived eccDNAs. eQTL information was retrieved from the PanCanQTL database [31], which includes survival-related, *cis*-eQTL, and trans-eQTL annotations across multiple cancer types. In this study, we focused on survival-related eQTLs to assess their enrichment within eccDNA regions. CTCF binding site coordinates were obtained from the CTCFBSDB database [32], which catalogs experimentally determined CTCF binding sites in various cancer cell lines. These were used to explore the potential association between eccDNAs and chromatin structural regulation. Methylation site data were sourced from the RMBase database [33], including known modification types such as m1A, m5C, m6A, and m7G. Coordinates of microRNA (miRNA) loci were retrieved from the miRBase database [34] to assess whether eccDNAs overlapped with regions involved in miRNA biogenesis. Annotations of repetitive elements, including LINEs, SINEs, LTRs, and satellite sequences, were compiled from the UCSC Genome Browser dataset. Finally, annotation files for protein-coding and non-coding genes were obtained from the Ensembl human hg38 gene annotation release and were used in the overlap and enrichment analyses to examine the relationship between eccDNAs and functional genomic elements.

Training and control data preparation

As deep learning models require both positive and negative examples for effective learning, we constructed matched positive and negative datasets for the two-stage DeepECC model training. In the first stage, DeepECC was trained to determine whether an input sequence corresponds to an eccDNA start or end breakpoint. Positive samples were derived from the genomic sequences flanking known eccDNA breakpoints. Specifically, we extracted sequences of fixed length centered on the start or end coordinates directly from the reference genome. Multiple input lengths were tested during training to evaluate their impact on model performance. Corresponding negative samples were generated by randomly selecting genomic positions on the same chromosome as the positive samples, ensuring equal sample size (1:1 ratio). To minimize sequence similarity between true breakpoints and random sites and reduce potential noise, all negative sites were selected to be located outside the model input sequence regions of any known eccDNA start or end breakpoint. The same flanking sequence length was then extracted from these positions in the reference genome to construct the negative input set. In the second stage, DeepECC was trained to evaluate whether a pair of sequences, representing a putative start and end breakpoint, could form a real eccDNA region. For positive samples, we ex-

tracted the flanking sequences of known start–end pairs and concatenated them as model input. For negative samples, we employed a “pseudo-pairing” strategy, in which start and end sequences from non-matching breakpoints on the same chromosome were randomly combined. At this stage, the input sequence length remains consistent with the previous step. Since the flanking sequences of the start and end breakpoints are concatenated, each sequence is halved before concatenation to maintain the overall length. This approach simulates non-circular combinations, thereby improving the model’s ability to discriminate between true and false eccDNA structures.

For all DeepECC training tasks, the complete dataset was randomly split into an 8:2 ratio, with 80% used for training and 20% reserved as a test set to evaluate model performance.

To further support downstream analyses of predicted eccDNA breakpoints and regions, we constructed randomized control datasets for statistical comparison. For breakpoint-level controls, we randomly selected an equal number of genomic positions without applying any positional filtering, ensuring a high degree of randomness. For eccDNA region-level controls, which required continuous genomic segments, we used the shuffle function in Bedtools [35] to generate background regions that matched the true eccDNA regions in number and length distribution, while preserving chromosome length proportions. In both control datasets, we ensured that the number of samples per chromosome matched the distribution of the actual eccDNA data to avoid introducing chromosomal bias into downstream enrichment analyses.

Hard negative mining

To mitigate the impact of potential noise in the eccDNA dataset and improve model generalization, we did not aim for near-perfect prediction accuracy during DeepECC training. Given the low prevalence of true eccDNA breakpoints in the genome, the model’s practical utility in large-scale genomic scanning lies more in effectively controlling the false positive rate than in maximizing raw accuracy. Therefore, in the first-stage genome-wide breakpoint prediction task, our focus was on optimizing the model’s discriminative ability and robustness, rather than accuracy alone. To this end, we incorporated a hard negative mining strategy [36, 37] to suppress false positives and improve model reliability. As illustrated in [Supplementary Fig. S3A](#), the procedure was as follows. First, an initial DeepECC model was trained using the original set of positive samples and randomly selected negative samples. This model was then applied to a new batch of randomly sampled negative sequences, and for each sequence, the probability of being classified as a true eccDNA breakpoint was calculated. Because the randomly sampled negatives may contain unannotated true breakpoints (i.e. latent positives), sequences with predicted probabilities in the range of 0.5–0.8 were regarded as high-risk false positives. These high-probability false negatives were added to the negative training set, and the model was retrained. This iterative process continued until the proportion of samples within the 0.5–0.8 probability range in the new negative samples dropped below 2%, thereby significantly enhancing the model’s robustness against genomic noise. Furthermore, to improve the practical reliability of genome-wide breakpoint prediction, we adopted a conservative thresholding approach during inference. Only sites with a model-predicted probability ≥ 0.95 were classified as potential eccDNA breakpoints. This stringent thresholding

strategy enabled high specificity, reduced false positives, and improved the practical utility and credibility of DeepECC in genome-wide eccDNA breakpoint discovery.

Model architecture

In this study, we developed a deep learning model based on a CNN architecture. The design was inspired by models such as SpliceAI [38], which have demonstrated strong performance in sequence-based modeling tasks. The model uses residual blocks as its core building units, with multiple blocks stacked together to form the overall architecture. Two residual blocks constitute a residual unit, and skip connections are applied between units to improve gradient flow and enhance training stability. The final output from the residual modules is passed through a fully connected layer to predict whether an input sequence represents an eccDNA breakpoint or eccDNA circle. The overall model structure is illustrated in Fig. 1A and Supplementary Fig. S1E.

Each residual block contains two 1D convolutional layers, each coupled with batch normalization and LeakyReLU activation. Skip connections are employed within the blocks to facilitate feature propagation. The key hyperparameters for each residual block include the number of filters ($N = 32$), kernel size ($W = 11$), and dilation rate ($D = 1$). We evaluated architectures with 1–4 stacked residual units to balance classification accuracy and computational efficiency. The final model employs two residual units, which achieved the best trade-off between performance and resource usage. Following the residual module, we included an additional 1D convolutional layer to further aggregate features. Subsequently, the model applies two stages of max-pooling (with pooling window sizes of 5 and 4, respectively) to progressively reduce the feature map dimensionality while retaining contextual information. The resulting compressed feature map is flattened and passed through a three-layer fully connected neural network for final classification. Each dense layer includes a linear transformation, LeakyReLU activation, and dropout (with a dropout rate of 0.2) to enhance expressiveness and reduce overfitting. The final output layer uses a sigmoid activation function to produce the classification probability for the binary prediction task.

The model was implemented in PyTorch version 2.3.1 and executed on the Intel Xeon Gold 6226R CPU with three RTX 4090 24G GPUs. Training was conducted using the AdamW optimizer with a learning rate of 0.0001 in the first stage and 0.001 in the second stage, and a weight decay of 0.001 for regularization. The number of training epochs was set to 30, and the batch size was fixed at 1024. For input encoding, we extracted genomic sequences of fixed length centered around each breakpoint and its corresponding negative sample from the reference genome. The sequences were transformed into a neural network-compatible format using one-hot encoding. Specifically, the four standard nucleotides A, C, G, and T were encoded as [1,0,0,0], [0,1,0,0], [0,0,1,0], and [0,0,0,1], respectively. Non-standard bases such as N were encoded as [0,0,0,0].

Model evaluation metrics

To comprehensively evaluate the classification performance of our deep learning model, we adopted multiple standard metrics, including accuracy, recall, precision, F1-score, and the area under the ROC curve (AUC). $\text{Accuracy} = (\text{TP} + \text{TN})$

$/ (\text{TP} + \text{TN} + \text{FP} + \text{FN})$, where TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively. $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$. $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$. $\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$. In addition to the above, we also calculated the area under the ROC curve (AUC) to assess the overall discriminative capability of the model. The ROC curve plots the true positive rate ($\text{TPR} = \text{TP} / [\text{TP} + \text{FN}]$) against the false positive rate ($\text{FPR} = \text{FP} / [\text{FP} + \text{TN}]$). These evaluation metrics jointly provide a quantitative basis for assessing the model's effectiveness in both eccDNA breakpoint identification and region-level prediction tasks.

Mutation sensitivity analysis

To investigate the sensitivity of the trained model to local base changes within input sequences, we conducted a systematic single-nucleotide mutation analysis. Specifically, for each input sequence, we performed a position-wise mutation by substituting the original base at each position with the other three possible nucleotides, one at a time, generating a set of single-point mutated sequences. Each of these mutated sequences was then passed through the trained DeepECC model to obtain a new prediction score. The impact of a mutation at a given position was quantified by calculating the difference between the model's output for the original (wild-type) sequence and that for the mutated sequence. For a sequence of length L , a total of $3 \times L$ mutated variants were generated, covering all possible base substitutions at each position. After performing this mutation scan across all sequences in the dataset, we aggregated the results to compute the average effect of each of the 12 possible base substitution types (e.g. $A \rightarrow C$, $A \rightarrow G$, $A \rightarrow T$, etc.) at different positions within the input. This analysis enabled us to identify specific sequence positions and nucleotide combinations that have a greater influence on the model's predictions, thereby uncovering the model's base-level sensitivity patterns and contributing to a better understanding of its learned sequence representations.

Regional scoring and overlap analysis

In the second stage of our study, we used the DeepECC model to assign combinatorial scores to all predicted pairs of eccDNA start and end breakpoints. Specifically, given that most eccDNAs are relatively short ($< 10\,000$ bp), for each candidate start site, we searched within a window ranging from 50 to 10 000 bp to identify all potential end sites. This window size captures most plausible eccDNA lengths while controlling the number of start–end pairs to maintain computational feasibility. Each start–end pair was then evaluated by the DeepECC model to predict the probability of forming a valid eccDNA junction. Pairs with a predicted probability > 0.5 were considered candidate eccDNA breakpoint combinations.

For each such pair predicted to form a potential eccDNA, we assigned the predicted probability of that pair to all nucleotide positions within the corresponding genomic interval between the start and end sites. Since a single base position may be covered by multiple breakpoint combinations, we accumulated the scores of all overlapping combinations at each base. To normalize the values and enhance interpretability, cumulative scores were transformed using a $\log(\text{score} + 1)$ scale. Base positions not covered by any predicted eccDNA pairs were assigned a score of zero. The resulting cumulative value

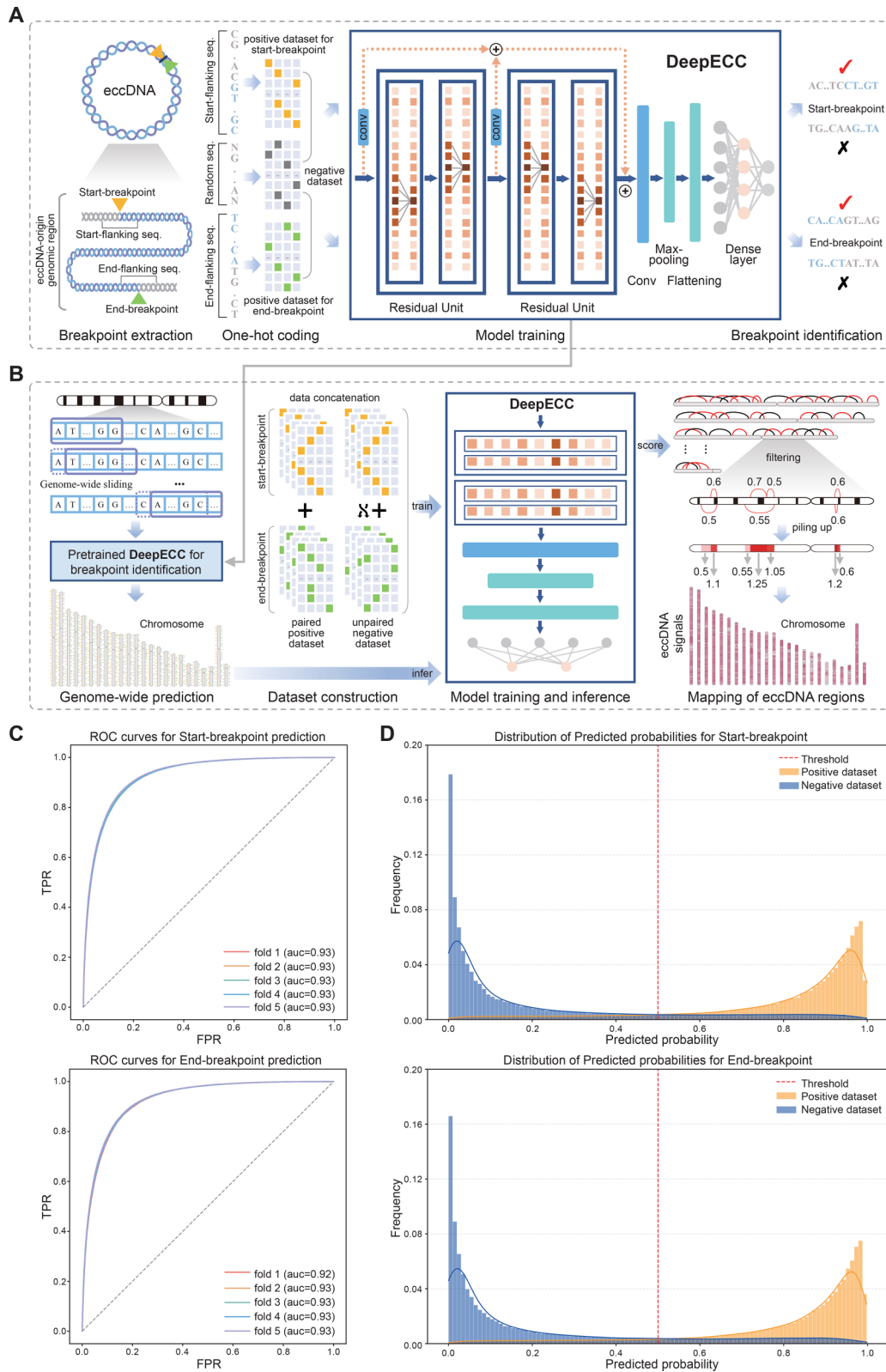


Figure 1. Framework for eccDNA identification and performance evaluation of the DeepECC model. **(A)** Stage I of the eccDNA identification framework: start and end breakpoints are extracted from known eccDNAs in the database. The sequences are one-hot encoded and input into a residual CNN for training, yielding two models for start and end breakpoint prediction. **(B)** Stage II of the framework: the breakpoint prediction models from Stage I are used to identify potential breakpoints genome-wide. Predicted breakpoints are then paired to generate candidate eccDNA regions, and a model with the same architecture scores the paired start–end breakpoints to obtain genome-wide eccDNA signals. **(C)** Receiver operating characteristic (ROC) curves from five-fold cross-validation on the start (top) and end (bottom) breakpoint datasets. **(D)** Predicted probability distributions for positive and negative samples in the start (top) and end (bottom) breakpoint datasets.

was defined as the eccDNA signal, indicating the likelihood of eccDNA formation at the corresponding nucleotide position.

Using this genome-wide, base-resolution eccDNA probability map, we further assessed the overlap between predicted eccDNA regions and functional genomic features. To achieve this, we used the ComputeMatrix module from Bedtools to calculate the potential eccDNA signal distribution within annotated genes and their ± 10 kb flanking regions. In addition, we used the Intersect module to quantify the overlaps between eccDNA regions and other genomic features, and subsequently calculated the proportion of overlap. The statistical significance of the overlaps between eccDNA regions and multiple genomic features was assessed using the chi-square test. To control for false positives arising from multiple testing, *P*-values were adjusted using the Benjamini–Hochberg method.

Gene pathway enrichment analysis

To investigate the potential biological functions of genes associated with predicted eccDNA regions, we performed Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analyses using the clusterProfiler package. Given the widespread genomic distribution of eccDNAs, we aimed to improve the specificity and representativeness of the enrichment analysis by focusing on the top 10% of eccDNA regions with the highest scores across the genome (Supplementary Fig. S4C). Genes overlapping these high-scoring regions were selected for downstream functional annotation. For both GO and KEGG analyses, we applied a significance threshold of false discovery rate < 0.05 to identify statistically enriched biological processes and pathways.

Motif discovery associated with eccDNA regions

To identify sequence motifs potentially associated with eccDNA regions, we performed motif enrichment analysis using the MEME Suite [39] on sequences from model-predicted eccDNA regions. Given the irregular genomic distribution of predicted eccDNAs (see Fig. 4A) and the known variability in eccDNA lengths observed in cells, we partitioned all predicted eccDNA regions into non-overlapping 500 bp windows. This segmentation yielded shorter eccDNA sub-fragments, each ≤ 500 bp in length, which were used as input for motif discovery. Although this approach may truncate longer eccDNA sequences, the resulting fragments better conform to the input length requirements of MEME, thereby facilitating motif detection. During motif search, the minimum and maximum motif widths were set to 6 and 25 bp, respectively. We extracted the top eight motifs with the highest statistical significance for downstream functional annotation and pathway enrichment analysis [40].

Results

Overview of the DeepECC framework for genome-wide eccDNA identification

To overcome the wide length heterogeneity of eccDNAs, fully leverage the sequences flanking their breakpoints, and enable flexible genome-wide eccDNA identification, we developed DeepECC, a deep learning framework for sequence-based eccDNA identification from the whole genome. DeepECC features a two-stage detection framework, with the first stage for breakpoint scanning (Fig. 1A) and the second stage to predict

the likelihood of any given pair of start–end breakpoints potentially forming a closed circular structure (Fig. 1B).

In the first stage, a CNN-based deep learning architecture was designed to identify eccDNA breakpoints from genomic sequences. For model training, eccDNA breakpoint data from cancer cell lines were obtained from the CircleBase database [24]. Positive samples were defined as genomic sequences centered on known start and end breakpoints, while negative samples of the same size were generated by randomly extracting sequences of the same length from the human reference genome. Each input was represented by the flanking regions surrounding a breakpoint site, encoded into fixed-length sequences using one-hot representation (Fig. 1A). Given the biological complexity of eccDNA formation, not all breakpoint sites are concentrated in a few highly recurrent genomic regions. Some breakpoint annotations from databases may contain biological noise or non-canonical events. Therefore, to prevent overfitting to spurious patterns, we deliberately avoided excessively long input sequences and deep neural architectures. To determine the optimal input length, we systematically evaluated the model on multiple sequence lengths. As shown in Supplementary Fig. S1A, B, and C, computational cost for training increased steadily with longer input sequences (Supplementary Fig. S1C), whereas performances on either the training (Supplementary Fig. S1A) or testing (Supplementary Fig. S1B) datasets showed no substantial improvement beyond 1500 base pairs. Therefore, 1500 bp was chosen as the input length, striking a balance between performance gain and computational efficiency. We then compared architectures of varying depth and selected a two-layer residual unit (RU-2) design to ensure that the model primarily captures generalizable sequence features of eccDNA breakpoints rather than overfitting to outliers or artifacts (Supplementary Fig. S1D). The overall performance of the final model was evaluated using five-fold cross-validation. The model consistently achieved robust performance, with an average ROC AUC of ~ 0.93 on both the start and end breakpoint prediction (Fig. 1C). The predicted probability distributions further demonstrated clear separation of positive and negative samples (Fig. 1D), highlighting the strong discriminative power of the learned features. Finally, the genome-wide scanning pretrained DeepECC model for potential eccDNA breakpoint identification was conducted using 1500 bp input sequences and a two-layer residual CNN architecture.

In the second stage of the workflow (Fig. 1B), we first employ the pretrained DeepECC model for breakpoint identification to perform genome-wide scanning, thereby obtaining all start and end breakpoints. Subsequently, we used pairwise start–end breakpoints from the database to train another classification model with the same architecture used in stage one. The framework operates effectively with a concatenated input length of 1500 bp for paired start–end breakpoints (Supplementary Fig. S4H), and in the second stage, model uses two 750 bp sequences centered on the start and end breakpoints as input. This model evaluates whether any given pair of start–end breakpoints could potentially form a closed circular structure. Finally, based on the candidate start and end breakpoints predicted, the model scores and integrates their pairwise likelihoods to construct a probabilistic landscape of eccDNA formation across the genome with single-base resolution. Although this method does not directly pinpoint the exact positions where eccDNAs are formed, it provides a powerful and flexible framework for modeling the po-

tential eccDNA-forming regions. This scoring-based approach supports downstream investigations into the mechanisms and distribution patterns of eccDNA, and is particularly suitable for qualitative analysis of eccDNA formation hotspots at the genome scale.

DeepECC robustly captures the eccDNA breakpoint features conserved across chromosomes and species

To investigate whether DeepECC can robustly capture conserved features flanking eccDNA breakpoints, we systematically analyzed multiple datasets from public resources. As shown in Fig. 1C, DeepECC demonstrated strong performance in recognizing eccDNA breakpoints, indicating that these regions contain informative and learnable sequence features. We next asked whether such features are shared across different chromosomes by the leave-one-chromosome-out experiment. We excluded eccDNA data from one chromosome at a time, trained DeepECC on the remaining data, and evaluated its ability to identify breakpoints on the excluded chromosome. As shown in [Supplementary Fig. S2A](#), the model retained high performance during training despite the exclusion of individual chromosomes. Moreover, when applied to detect breakpoints from the excluded chromosomes, DeepECC consistently achieved high classification accuracy, with most chromosomes reaching 0.8 and several exceeding 0.9 (Fig. 2A). In contrast, its accuracy on randomly sampled genomic positions was below 0.2, indicating that eccDNA breakpoints are enriched with shared flanking sequence rather than being randomly distributed across the genome.

To explore the performance and conservation across species, we incorporated additional datasets from the eccDNA Atlas [25] and TeCD [26], which cover eccDNAs from different conditions and species, including human cancer and normal samples, mouse, and gallus ([Supplementary Table S1](#)). We then trained separate DeepECC models on eccDNA breakpoints from different datasets and evaluated their prediction performance. As shown in [Supplementary Fig. S2B](#), DeepECC performed well on human cancer and mouse datasets, but less effectively on eccDNAs from human normal and gallus eccDNAs. This discrepancy is consistent with their breakpoint distribution ([Supplementary Fig. S2C](#)), where eccDNAs from human cancer and mouse exhibit more clustered breakpoints, whereas eccDNAs from human normal samples and gallus display broader distributions and more complex sequence structures, making it more difficult for the model to learn generalized features. To further examine whether the learned sequence features are transferable across species, we conducted cross-species prediction experiments. Specifically, models trained on human cancer and mouse datasets achieved strong accuracy when applied to each other's datasets, significantly outperforming random baselines (Fig. 2B). These results highlight that DeepECC effectively captures conserved breakpoint features shared across species, while also reflecting species-specific patterns.

Finally, to investigate the model's sequence preferences, we performed a base substitution analysis using the model trained on the CircleBase human cancer dataset. Each nucleotide position in the training sequences was systematically mutated to evaluate how specific base changes affect model predictions. As shown in Fig. 2C, DeepECC preferentially focused on flanking sequences within the eccDNA body between break-

point pairs than on sequences outside the eccDNA region, indicating that internal sequences contribute more to breakpoint recognition. Furthermore, substitutions from A/T to G/C generally increased the predicted probability of a breakpoint, whereas G/C to A/T mutations tended to decrease it. These findings suggest that DeepECC preferentially identifies breakpoints in GC-rich regions, and that the internal sequence context of eccDNA plays a more important role in model decision-making than external flanking genomic regions. This supports the hypothesis that eccDNA formation may be associated with GC content and local sequence stability.

Together, these results demonstrate that the flanking regions of eccDNA breakpoints encode stable and learnable features. DeepECC not only maintains robust predictive power across different chromosomes but also generalizes these patterns across species, with particularly strong consistency in human cancer and mouse datasets. Combined with the base substitution analysis revealing a preference for GC-rich contexts, these findings indicate that eccDNA breakpoints harbor conserved flanking features that DeepECC can reliably capture, providing new insights into the mechanisms of eccDNA formation and its cross-species conservation.

DeepECC enables genome-wide analysis of eccDNA breakpoints with varying genomic enrichment

By training the DeepECC model, we achieved accurate identification of both the start and end breakpoints of eccDNAs directly from DNA sequences. Leveraging this capability, we applied DeepECC to perform a genome-wide scan of the human reference genome to identify potential eccDNA breakpoints. To minimize false positives and improve precision, we incorporated a hard negative mining strategy ([Supplementary Fig. S3A](#)). Through eight rounds of iterative hard negative mining, models progressively improved and ultimately satisfied the predefined threshold ([Supplementary Fig. S3C](#)). The genome-wide distribution of candidate breakpoints identified by DeepECC is shown in Fig. 3A. These predicted breakpoints are non-uniformly distributed across chromosomes, with chromosomes 17 and 19 exhibiting notably higher densities, consistent with previous observations [41]. Given that these chromosomes harbor higher gene densities and that genic regions are generally enriched in GC content, we hypothesize that the elevated GC content contributes to the observed increase in breakpoint density. Further analysis confirmed that DeepECC-predicted breakpoints are preferentially located in GC-rich regions (Fig. 3B), consistent with the base substitution analysis in Fig. 2C, where A/T to G/C changes increased predicted breakpoint probabilities. Together, these results indicate that DeepECC effectively learned sequence features associated with GC-enriched regions in eccDNA breakpoint prediction, particularly in human cancer cell data.

To explore whether these predicted breakpoints preferentially localize to specific genomic features, we performed a statistical intersection analysis between the predicted breakpoints and 38 annotated genomic feature categories that are grouped into four categories, including 13 types of repetitive regions, 5 structural variations, 15 chromatin states, and 5 regulatory regions. As shown in Fig. 3C and [Supplementary Fig. S3C](#), the overlap counts between predicted breakpoints and these genomic features were significantly different from those

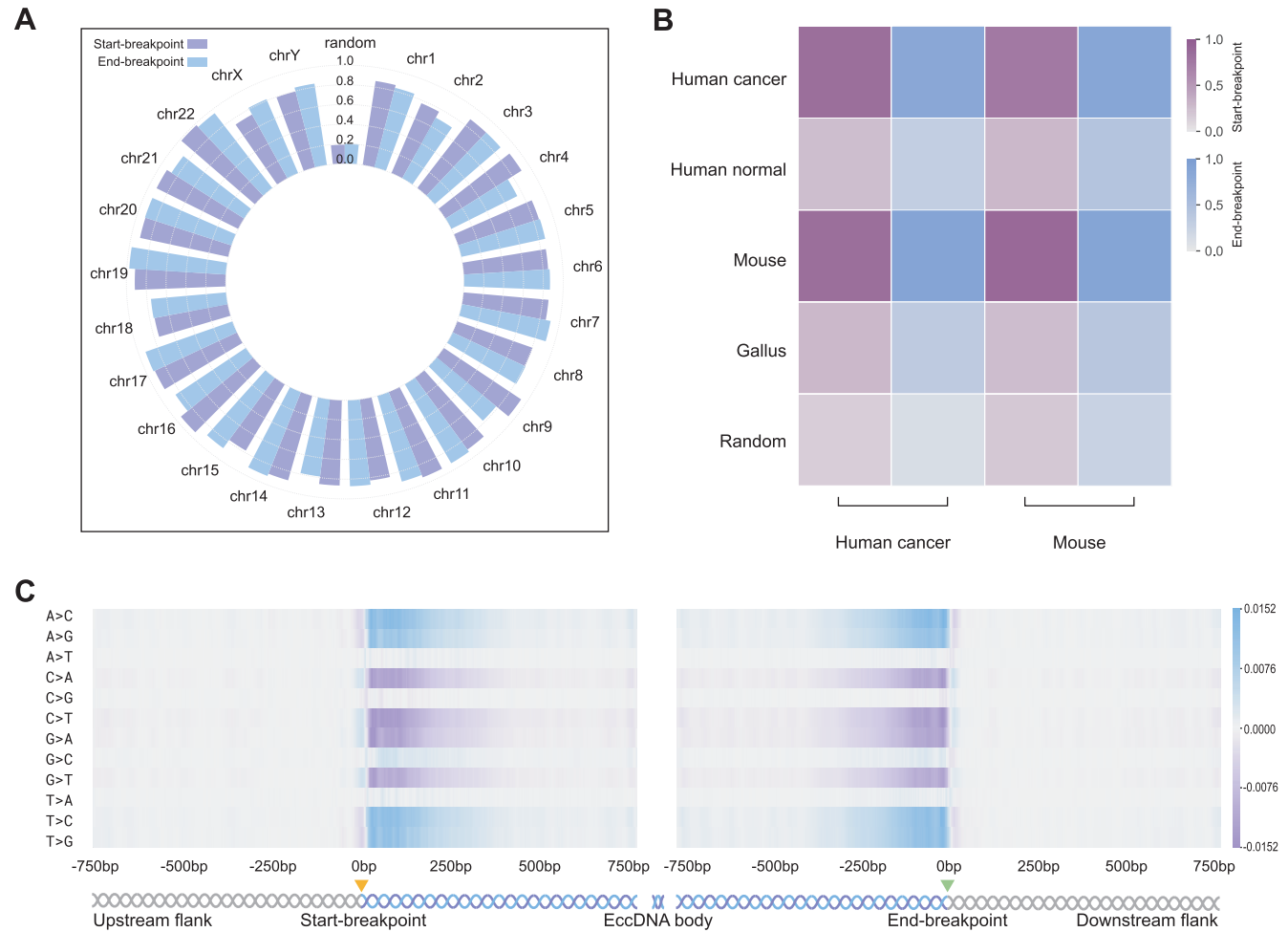


Figure 2. Evaluation of model generalizability and sequence preference. **(A)** Leave-one-chromosome-out validation results for start and end breakpoint prediction. Each bar indicates prediction performance on the excluded chromosome using models trained on the remaining chromosomes. **(B)** The heatmap shows the prediction performance when models are trained (x-axis) and tested (y-axis) on cross-species datasets, including human cancer, human normal, mouse, and gallus eccDNAs, as well as random sequences. **(C)** Results of base substitution perturbation. The left and right panels show sequences surrounding the start and end breakpoints, respectively.

obtained from randomly sampled genomic loci, with full statistical details summarized in [Supplementary Table S2](#) and [S3](#). Specifically, within the repeat regions, only five features, including G-quadruplex-forming sequences and Z-DNA motifs, showed significant enrichment at breakpoint sites, while the remaining eight repeat regions exhibited lower-than-expected overlap. Among structural variation regions, only INS regions displayed higher-than-expected overlap, suggesting that eccDNA breakpoints are generally not enriched in regions prone to structural genomic rearrangements. In contrast, chromatin state analysis revealed significant enrichment of eccDNA breakpoints in active chromatin, particularly active transcription start sites (TssA) and their flanking regions, as well as enhancers, whereas heterochromatin (Het) and quiescent (Quies) regions showed reduced overlap. Furthermore, regulatory regions analysis demonstrated significant enrichment at CpG islands, DNA methylation sites, and promoter regions, consistent with a tendency for eccDNA to arise from functionally important genomic loci.

Collectively, these findings suggest that eccDNA formation is primarily driven by disruptions in transcriptionally active, GC-rich, and functionally relevant genomic regions,

rather than by structural chromosomal abnormalities. This aligns with current hypotheses that eccDNAs are generated as byproducts of aberrant DNA replication or transcriptional processes [4]. Overall, DeepECC provides a robust framework for genome-wide identification of eccDNA breakpoints, revealing their preferential formation in regions with high GC content, active chromatin, and regulatory importance, thereby highlighting the potential mechanistic links between eccDNA generation and genomic stress or functional perturbations.

DeepECC uncovers eccDNA hotspots with potential functional and pathogenic relevance

After obtaining all candidate breakpoint sites, we proceeded to the eccDNA circularization identification workflow (as shown in Fig. 1B), wherein all possible combinations of predicted start–end breakpoints were evaluated. First, we trained a DeepECC model to classify breakpoint pairs based on the true start–end pairs extracted from the CircleBase human cancer eccDNA dataset. As shown in [Supplementary Fig. S4A](#) and [B](#), the model achieved high classification performance, with AUC scores exceeding 0.9 on both the training and test-

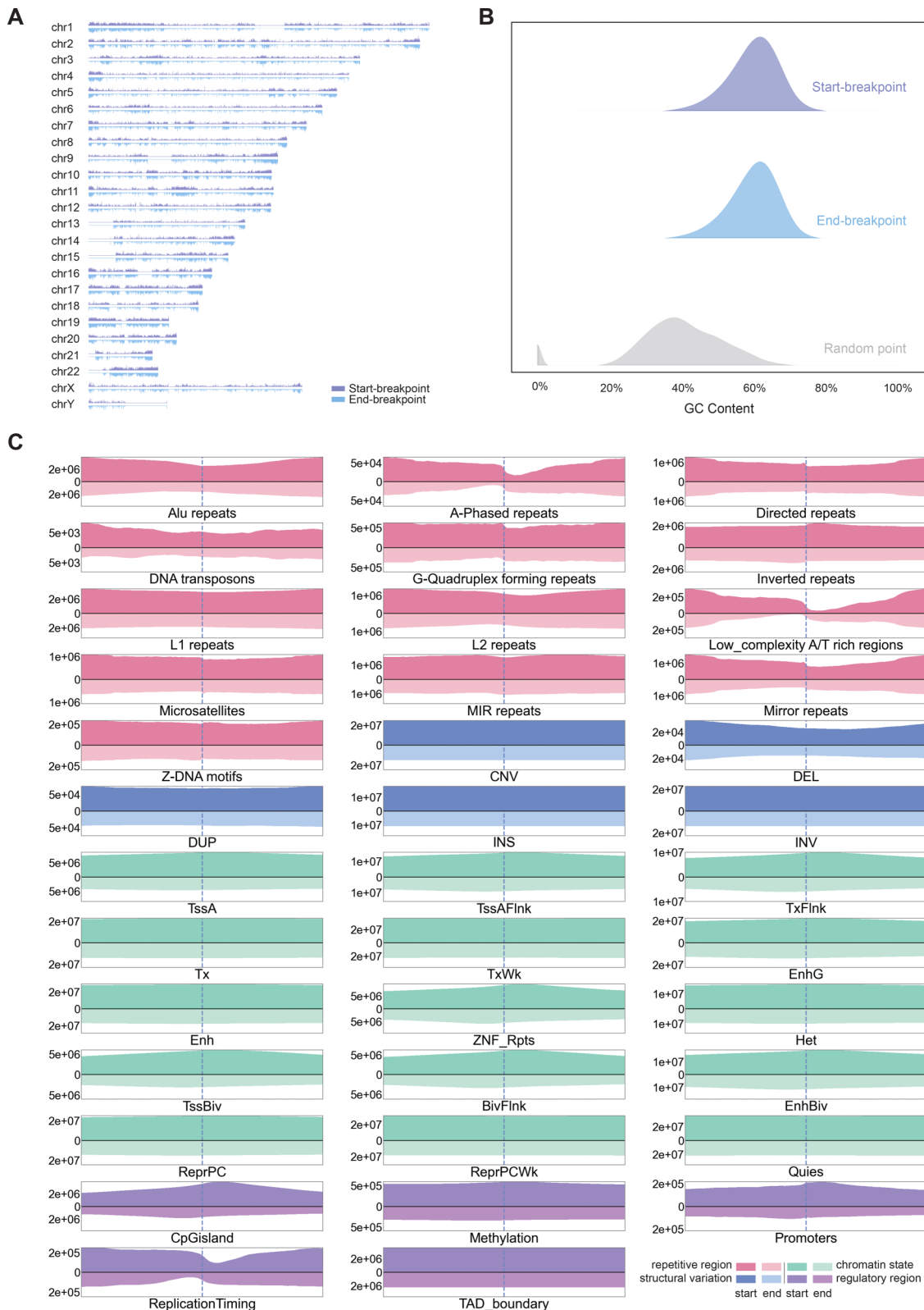


Figure 3. Genomic distribution and feature associations of predicted eccDNA breakpoints. **(A)** Genome-wide distribution of predicted start and end breakpoints across all 24 human chromosomes. **(B)** GC content distribution of sequences surrounding the predicted start and end breakpoints compared to randomly selected genomic points. **(C)** Genomic feature enrichment profiles centered on predicted start and end breakpoints. Features are grouped into four categories: repetitive elements, structural variations, chromatin states, and regulatory elements. Start and end breakpoint signals are shown separately for each feature.

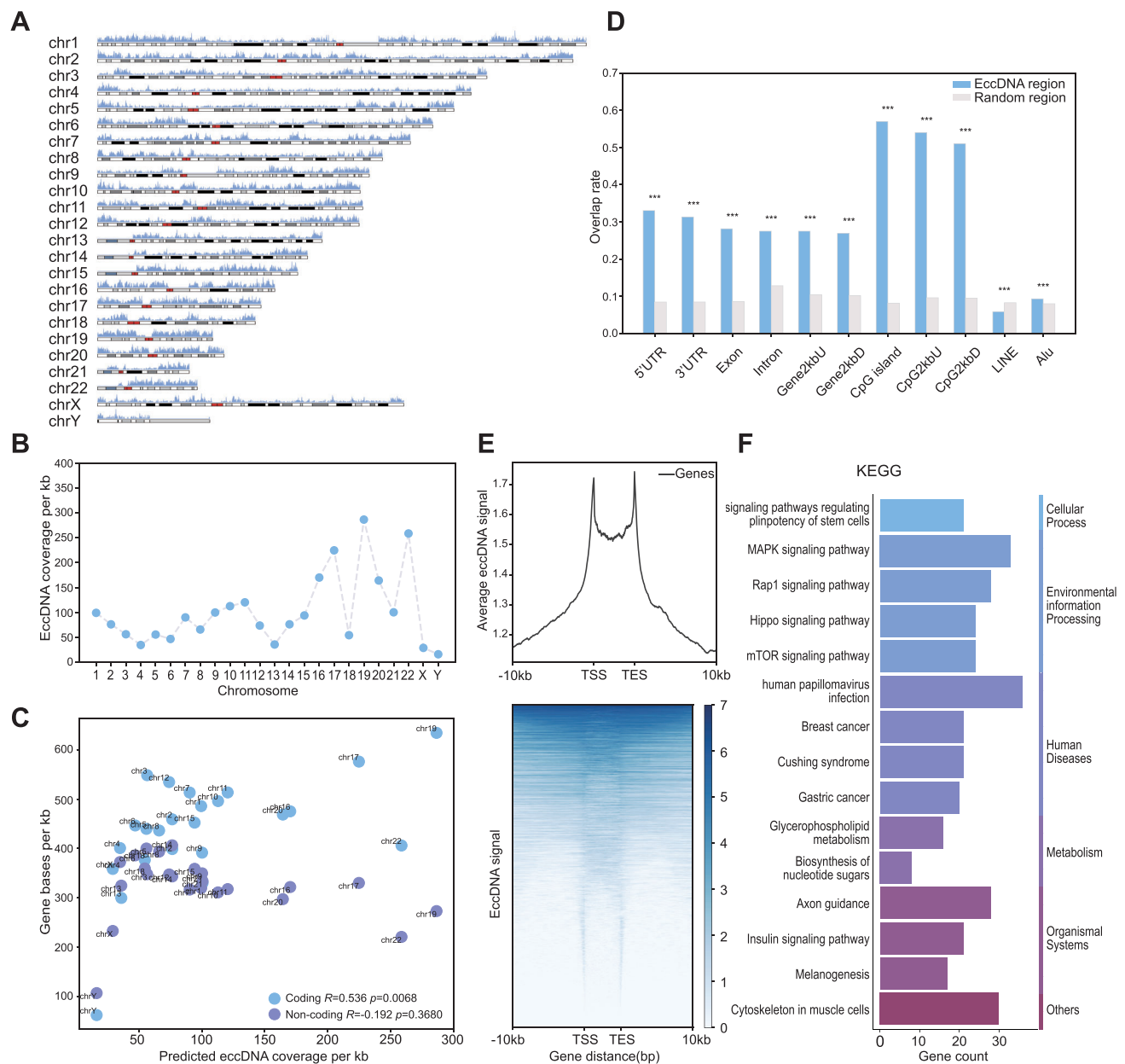


Figure 4. Genomic distribution of eccDNA regions reveals hotspots with potential functional and pathogenic relevance. **(A)** Genome-wide distribution of predicted eccDNA scores on human chromosomes, visualized together with chromosomal ideograms. **(B)** The eccDNA coverage per kilobase for each chromosome. **(C)** Correlation between eccDNA coverage and gene density across chromosomes, separately shown for coding and non-coding genes. **(D)** The overlap rates between eccDNA regions and various genomic elements, including untranslated regions (UTRs), introns, gene flanking regions, CpG islands, and repetitive elements. *** P -value $< .001$. **(E)** Profile plot (top) and heatmap (bottom) of eccDNA scores around gene regions, centered on transcription start sites (TSSs) and transcription end sites (TESs), generated by ComputeMatrix. **(F)** KEGG pathway enrichment of genes overlapped with the top 10% highest-scoring predicted eccDNA regions.

ing datasets. Using this trained model, we performed pairwise predictions across all candidate breakpoints to generate a genome-wide eccDNA signal map, where each genomic base was assigned a score reflecting its likelihood of being part of an eccDNA. As shown in Fig. 4A, the distribution of predicted eccDNA signals across the genome is clearly non-uniform. When compared with the chromosomal ideogram, we observed that centromeric regions (marked in red) generally displayed low eccDNA scores, consistent with previous findings that eccDNAs rarely originate from centromeres [2]. This supports the biological validity of the model's predictions. Simi-

larly, heterochromatin regions (marked in black) also exhibited lower scores, while euchromatic regions (marked in light gray) showed relatively higher scores, suggesting that eccDNAs are more likely to arise from active genomic regions.

To quantify eccDNA density, we calculated the average number of eccDNA-covered bases per 1 kb window across each chromosome. The results showed that chromosomes with high gene density, such as Chr19, Chr17, Chr22, and Chr16, also displayed elevated eccDNA base coverage (Fig. 4B). In contrast, gene-poor chromosomes, including ChrY, Chr13, Chr18, Chr21, and ChrX, exhibited

significantly lower eccDNA coverage. To further investigate the relationship between eccDNA regions and gene distribution, we extracted the locations of protein-coding and non-coding RNA genes from the reference genome annotation and performed correlation analyses with the eccDNA density (Fig. 4C). The results revealed a positive correlation between coding gene density and eccDNA signal across chromosomes, particularly evident in Chr19 and Chr17, which are both gene-rich and eccDNA-enriched.

We next assessed the overlap between eccDNA regions and key genic elements such as exons. As shown in Fig. 4D and [Supplementary Table S4](#), the predicted eccDNA regions are significantly enriched in gene bodies and their proximal flanking regions, as well as transcription regulatory elements, but depleted in canonical repeats (e.g. LINE and Alu), suggesting highly functional relevance of predicted eccDNAs. To comprehensively assess eccDNA distribution within gene bodies, we employed ComputeMatrix to compute eccDNA scores across all annotated gene regions. As illustrated in Fig. 4E, eccDNA signals were significantly elevated within gene bodies and proximal flanking regions compared to distal flanking upstream and downstream regions. Notably, the highest scores were observed near TSSs and TESs. These findings further support the conclusion that eccDNAs preferentially arise within functionally active genic regions. To further characterize the functional landscape of genes enriched within eccDNA hotspots, we extracted the top 10% high-scoring eccDNA hotspot regions ([Supplementary Fig. S4C](#)) and identified overlapping protein-coding genes within these regions ([Supplementary Table S5](#)). GO and KEGG pathway enrichment analyses (Fig. 4F and [Supplementary Fig. S4D–F](#)) revealed significant enrichment in several key signaling pathways, including MAPK and Rap1, as well as disease-associated pathways such as breast cancer and gastric cancer.

To facilitate exploration of the prediction results, we integrated the genome-wide eccDNA identified by DeepECC into the UCSC Genome Browser (https://genome.ucsc.edu/s/ccbioinfo/EccDNA_Hub), enabling researchers to query and visualize whether their genes or regions of interest overlap with cancer-associated eccDNA. This genomic annotation provides a novel layer of functional information, supporting experimental scientists in rapidly assessing the potential role of eccDNA formation in their studies. Importantly, examination of several well-documented genes known to play critical roles in tumorigenesis through eccDNA activity, including CCND1, EGFR, MYC, and MYCN [42–44], is in line with previously reported experimental observations ([Supplementary Fig. S4G](#)). This demonstrates both the reliability of our predictions and the utility of the browser-based resource for cancer research.

Taken together, these results demonstrate that DeepECC effectively captures the genome-wide distribution of eccDNA signals and reveals their strong preference for transcriptionally active, gene-dense, and functionally important regions. By integrating base-level scoring, density quantification, and functional enrichment analyses, DeepECC highlights eccDNA hotspots that are not only enriched near key regulatory elements but are also associated with critical signaling and disease pathways. These findings underscore the biological relevance of DeepECC predictions and suggest that eccDNA formation is closely linked to genomic activity and cellular functional states.

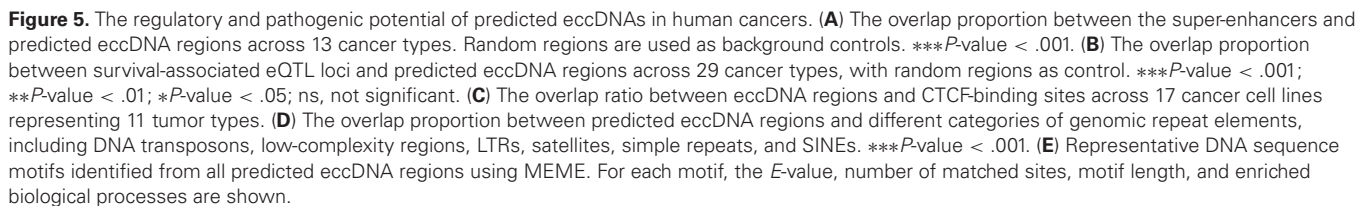
DeepECC highlights the regulatory and pathogenic potential of predicted eccDNAs in human cancers

The observation that genome-wide eccDNA hotspots are enriched within gene bodies and regulatory regions motivated us to further investigate their potential roles in cancers. To this end, we systematically analyzed the overlap of predicted eccDNA regions with a wide range of functional genomic elements curated from multiple databases, including enhancers, eQTLs, CTCF binding sites, repeat elements, and DNA methylation sites. Notably, the enhancer, eQTL, and CTCF datasets span multiple cancer types, and have been shown to exert varying degrees of regulatory influence on gene expression in tumor cells.

Firstly, we observed significantly higher overlap of eccDNAs with both super enhancers and typical enhancers across 13 cancer types compared to randomly generated genomic regions (Fig. 5A, [Supplementary Fig. S5A](#), and [Supplementary Table S6](#)), suggesting that eccDNAs may carry enhancer fragments and potentially contribute to transcriptional regulation in tumor cells. In the eQTL analysis, eccDNA regions were significantly enriched for survival-associated eQTLs in several cancers, including BLCA, GBM, STAD, and UVM (Fig. 5B and [Supplementary Table S7](#)), indicating that eccDNAs may affect tumor progression and patient survival by incorporating regulatory elements to influence gene expression. However, enrichment patterns were not uniform across all cancer types, suggesting that eccDNA-mediated eQTL effects are cancer-type specific or context-dependent. Analysis of CTCF binding sites across 11 cancer types revealed that eccDNA regions consistently showed higher overlap with CTCF sites compared to random controls (Fig. 5C and [Supplementary Table S8](#)), implying potential involvement of eccDNAs in higher-order chromatin organization through CTCF-mediated interactions. Additionally, eccDNA were notably enriched at certain DNA methylation sites and miRNA loci ([Supplementary Fig. S5B and C](#); [Supplementary Tables S10 and S11](#)), further supporting their origin from functionally important regulatory regions of the genome. In contrast, canonical repeat regions associated with chromosomal stability, such as LINEs, SINEs, LTRs, and satellite elements, showed little to no enrichment, and in some cases were even underrepresented (Fig. 4E and 5D; [Supplementary Table S9](#)), reinforcing the notion that eccDNAs preferentially form in biologically active and regulatory genomic domains rather than structurally repetitive regions.

To gain insight into the sequence-specific features of eccDNA regions, we performed motif discovery using the MEME suite. The top eight significantly enriched motifs (Fig. 5E) were widely distributed across predicted eccDNA regions. Functional annotation of the associated transcription factors revealed enrichment in biological processes such as ribosome biogenesis, translational regulation, RNA metabolism, and the DNA damage response. These results suggest that the sequence content of eccDNAs may play a role in protein synthesis, RNA processing, and genome stability.

Taken together, these analyses demonstrate that eccDNAs are predominantly derived from functionally critical regulatory regions of the genome, carrying enhancer elements, survival-associated eQTLs, and CTCF binding sites. The incorporation of these regulatory sequences likely endows eccDNAs with the capacity to modulate gene expression, influence chromatin architecture, and potentially alter cellular states.



In the context of cancer, such functional contributions may facilitate transcriptional reprogramming, promote oncogenic gene expression, and impact disease progression and patient prognosis, highlighting eccDNAs as active players in genome regulation rather than passive byproducts of DNA rearrangements.

Discussion

In this study, we developed DeepECC, a deep learning framework designed to learn the intrinsic sequence features surrounding eccDNA breakpoints. By focusing on breakpoint-flanking regions rather than eccDNA body sequences alone, DeepECC effectively addresses the limitations of previous fixed-length input deep learning approaches, such as DeepCircle, which struggle to generalize across eccDNAs of highly variable lengths. Moreover, by combining the predicted start and end breakpoints, DeepECC generates a continuous genome-wide probability landscape, allowing quantitative identification of eccDNA-forming hotspots. This scoring framework not only facilitates the identification of high-confidence eccDNA regions, but also provides biological insight into where and how eccDNAs are likely to form, especially in tumor genomes. Our findings reveal that eccDNAs exhibit strong positional preferences and are significantly enriched in gene-dense and functionally active genomic regions. These observations offer a mechanistic explanation for the complex functional roles that eccDNAs have been reported to play in cancer cells.

In the field of genomic functional region prediction, previous studies have shown that regions such as promoters and splice sites often possess conserved sequence features, including elements like the TATA box and the GT/AG dinucleotide motif, which are suitable for precise identification using deep learning models following appropriate data preprocessing and noise filtering [45, 46]. In contrast, the formation mechanisms of eccDNAs are more complex, and their internal sequences lack consistency, making it unclear whether breakpoints are predictable based on sequence alone. Our results demonstrate that, even under conditions where breakpoint formation may involve stochastic elements, the model can still effectively identify eccDNA breakpoints in cancer cells, supporting the notion that sequence regularities exist around breakpoint regions. Given the inherent noise and heterogeneity of eccDNA datasets, we prioritized model robustness and generalizability. Although advanced architectures, such as Transformer, are effective at capturing long-range sequence dependencies, their increased complexity may lead to overfitting in noisy eccDNA data. We therefore adopted a CNN-based architecture that provides a robust solution under current data conditions. As eccDNA datasets improve in scale and quality, integrating Transformer-based mechanisms represents a promising direction for modeling long-range dependencies. To address the challenge of limited interpretability inherent in deep learning models, we conducted an integrative analysis using multiple public databases to investigate the genomic distribution patterns of eccDNA breakpoints. Our findings show that eccDNA breakpoints are significantly enriched in GC-rich genomic regions, particularly near functional elements such as promoters. This observation is consistent with previous reports suggesting a preference for eccDNA formation at GC-rich loci [28]. Although the breakpoint-based prediction strategy does not cover the full internal sequence of eccDNAs, it

enables genome-wide mapping of potential eccDNA-forming regions. By combining predicted start and end breakpoints using the second-stage DeepECC model, we assigned scores to genomic intervals and constructed a comprehensive distribution landscape of eccDNA across the genome. Notably, we observed that eccDNA exhibits non-random chromosomal distribution, with significant enrichment on gene-dense chromosomes such as Chr17 and Chr19, in agreement with earlier findings based on sample-derived data [2]. Together, our two-stage deep learning framework provides a genome-wide, data-driven validation of prior experimental observations. Furthermore, we identified strong enrichment of eccDNA in protein-coding gene regions, enhancers, and regulatory elements such as transcription factor binding sites, as well as translation-associated motifs. These results suggest that eccDNAs may carry functionally relevant genomic fragments and potentially influence cellular states or enhance gene expression, offering novel insights into their roles in transcriptional regulation and disease processes.

During the course of this study, we observed that the model performed considerably better when trained on eccDNA data from human cancer cells or mouse tissues, while achieving limited success with human normal cell or gallus eccDNA datasets. This suggests that eccDNAs exhibit substantial heterogeneity across species and tissue types. Taking human cells as an example, eccDNAs are known to be widely present in normal cells, although there is currently no strong evidence to suggest that they play a significant functional role in normal cellular physiology. In contrast, numerous studies have demonstrated that eccDNAs in cancer cells exert diverse and biologically significant effects, indicating that the functional relevance and formation mechanisms of eccDNAs differ greatly across cell types. Our findings further suggest that eccDNAs tend to originate from functionally active genomic regions, potentially inheriting regulatory sequences that enable them to act as substitute enhancers or modulate transcription. Somewhat unexpectedly, we found that eccDNAs are not preferentially formed in regions associated with structural variation. It might seem intuitive that eccDNAs would arise from areas prone to chromosomal instability or breakage; however, our data show low correlation between predicted eccDNA regions and structural variant loci, suggesting that genome instability alone does not explain eccDNA genesis. Additionally, we observed opposing formation preferences of eccDNAs in protein-coding versus non-coding RNA gene regions. EccDNAs were more frequently formed in coding gene regions, whereas no evidence of association in non-coding gene regions. This further supports the hypothesis that eccDNA generation is closely associated with active transcription and that eccDNAs may preferentially arise from highly transcribed, protein-coding regions. This observation also helps explain why eccDNAs in cancer cells tend to exhibit enhanced transcriptional activity, as their genomic origins are often located in transcriptionally active chromatin.

Based on our current findings, eccDNAs in cancer cells exhibit distinct sequence and positional characteristics that can be effectively modeled using deep learning approaches. Our study demonstrates that the genome-wide distribution of eccDNA-forming regions is highly specific, aligning with previously observed functional features of eccDNAs. In particular, the enrichment of eccDNA in transcriptionally active, protein-coding regions provides a mechanistic explanation for their frequently observed high transcriptional activ-

ity and their potential to modulate gene expression in cancer cells. Moreover, our results strongly support the non-random nature of eccDNA formation, with significant enrichment at enhancers, CTCF binding sites, and eQTL loci. These findings help explain how eccDNAs may acquire enhancer-like regulatory functions and participate in transcriptional reprogramming processes that promote tumor progression. These conclusions provide a genome-wide, systems-level perspective on the formation and functional potential of eccDNAs, overcoming the limitations of sample-specific analyses. By revealing generalizable patterns of eccDNA biogenesis and distribution, our study offers a more comprehensive and biologically interpretable framework for understanding eccDNA function in cancer and potentially other disease contexts.

Although our deep learning model successfully predicted potential eccDNA-forming regions and revealed statistically significant distributional patterns, it is important to note that the model outputs probabilistic scores rather than direct evidence of eccDNA presence in cells. Accordingly, these predictions should be interpreted as reflecting relative formation tendencies and enrichment patterns, rather than definitive eccDNA events. Consequently, future studies are required to incorporate experimental validation to further elucidate the molecular mechanisms and functional consequences of eccDNA formation.

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Supplementary data

Supplementary data is available at NAR online.

Conflict of interest

None declared.

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Data availability

All data used in this study are obtained from publicly available databases, with detailed access and download information provided in [Supplementary Table S12](#). The DeepECC model and corresponding training code are available at <https://github.com/ccbioinfo/DeepECC> (<https://doi.org/10.6084/m9.figshare.30449276>). Additional scripts for data analysis and figure generation have been deposited to Figshare at <https://doi.org/10.6084/m9.figshare.30449276>.

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